

DATASETS, BENCHMARKS, AND PROTOCOLS

Assessment of Large Language Models in Clinical Reasoning: A Novel Benchmarking Study

Liam G. McCoy , M.D., M.Sc.,^{1,2,3} Rajiv Swamy , S.M.,⁴ Nidhish Sagar , S.M.,³ Minjia Wang , M.Eng.,⁴ Stephen Bacchi , M.B.B.S., Ph.D.,⁵ Jie Ming Nigel Fong , M.R.C.P.,⁶ Nigel C.K. Tan , M.S.-H.P.Ed., F.R.C.P.(Edin).,^{7,8} Kevin Tan , B.M.B.S., M.S.-H.P.Ed.,^{7,8} Thomas A. Buckley , B.S.,⁹ Peter Brodeur , M.D., M.A.,² Leo Anthony Celi , M.D., M.P.H., M.Sc.,^{2,3,10} Arjun K. Manrai , Ph.D.,^{9,11} Aloysius Humbert , M.D.,¹² and Adam Rodman , M.D., M.P.H.^{2,11}

Received: February 6, 2025; Revised: June 28, 2025; Accepted: July 11, 2025; Published: September 25, 2025

Abstract

BACKGROUND Large language models (LLMs) are increasingly deployed for clinical decision support, yet standard evaluations such as medical licensing exams overlook how clinicians update decisions in dynamic contexts. Script concordance testing (SCT), a decades-old assessment tool, measures how new information adjusts diagnostic or therapeutic judgments under uncertainty.

METHODS We built a public benchmark of 750 SCT questions drawn from 10 international, diverse datasets — 9 of which were newly released — spanning multiple specialties. Each item presents a clinical vignette and asks how added data change the likelihood of a diagnosis or management option, scored against expert-panel responses. Ten state-of-the-art LLMs were compared with 1070 medical students (1026 medical; 44 physiotherapy), 193 residents, and 300 attending physicians.

RESULTS LLMs demonstrated markedly lower performance on SCT than their typical achievement on medical multiple-choice benchmarks. Across prompting conditions, OpenAI's o3 achieved the highest performance (67.8%±1.2%), followed by GPT-4o (63.9%±1.3%). Other reasoning-optimized models, including OpenAI's o1-preview (58.2%±1.3%) and DeepSeek R1 (55.5%±1.4%), performed significantly lower, with Google's Gemini 2.5 Pro Preview (Gemini 2.5) (52.1%±1.4%) exhibiting the poorest results overall, while other non-reasoning models showed middling performance. Models matched or exceeded student performance on multiple examinations but did not reach the level of senior residents or attending physicians. Response-pattern analysis showed systematic overconfidence, with reasoning-tuned models overusing extreme ratings (+2/−2) and seldom choosing 0, implying that chain-of-thought optimizations may hinder flexible clinical reasoning under uncertainty.

CONCLUSIONS SCT exposes persistent limitations in LLM clinical reasoning, especially in models optimized for explicit reasoning. Although SCT performance offers analogies to human probabilistic adjustment, it represents only one facet of evaluating artificial intelligence decision support. Our human-validated benchmark, now publicly available, provides a rigorous tool for advancing the assessment of medical artificial intelligence systems. (Funded by the Fulbright Program and others.)

The author affiliations are listed at the end of the article.

Liam G. McCoy can be contacted at lmccoy@mit.edu.

Introduction

Large language models (LLMs) have been rapidly integrated into everyday clinical care, transcribing encounters,¹⁻³ drafting patient messages, summarizing clinical documentation,^{4,5} and providing point-of-care clinical decision support.^{6,7} With workforce shortages accelerating interest in LLM-augmented reasoning, responsible deployment demands robust, standardized evaluation. Although benchmarks cannot replace clinical trials, they have become indispensable early filters.

Measuring the performance of LLMs on reasoning tasks is difficult and controversial. Models perform exceptionally well on medical licensing exams and question-answering datasets, but these results likely reflect encoded knowledge and test-taking abilities rather than inherent reasoning.⁸ Similarly, models have shown increasingly good performance on historical and open-ended clinical cases, such as the *New England Journal of Medicine* clinicopathological conference series,⁹ but it is unclear how well these results translate to real-life clinical reasoning capabilities. Recent studies suggest that explicit reasoning traces may degrade model performance in tasks requiring flexible adjustment under uncertainty,¹⁰ underscoring the need for evaluation formats that stress these capacities.

Many measures of clinical reasoning adapted from medical education have been successfully used as benchmarks of model performance, including objective structured clinical examinations with patient actors,¹¹ structured reflection rubrics,¹² display of reasoning,¹³ and free-text multistep clinical vignettes.¹⁴ These evaluations have the advantage of decades of validity evidence¹⁵ and explicit comparisons with human clinicians, but they are also laborious to score and expensive to create, limiting their utility in large-scale assessments of models, especially given persistent risks of data leakage.

Script concordance testing (SCT) probes a core clinical reasoning skill — updating judgment in response to new information under uncertainty¹⁶ — yet remains trivially scalable. Each question begins with a clinical vignette containing elements of natural diagnostic or management ambiguity, similar to real-world practice (Fig. 1). Examinees are then presented with a potential intervention, diagnosis, or management option, followed by a new piece of clinical information. They must assess how this new information changes the likelihood of their initial hypothesis, quantified on a five-point scale from

“much less likely” to “much more likely.” Answers are scored against a reference panel of experienced clinicians, with maximum credit awarded for selecting the modal response, and partial credit for answers selected by at least some experts. The scoring mirrors legitimate expert disagreement in ambiguous cases.

We present the first openly available SCT benchmark for LLM reasoning (<https://www.concor.dance/>). The work pursues three goals: to assess how effectively LLMs can process and integrate new clinical information to adjust their diagnostic and therapeutic reasoning; to compare model performance with human medical trainees and practitioners at various levels of expertise; and to explore how different model architectures influence performance on these specialized reasoning tasks.

Although SCT captures important dimensions of diagnostic and management adjustment under uncertainty, it remains an intentionally constrained proxy and does not fully replicate real-world clinical reasoning. Nonetheless, the ability of SCT to reveal specific gaps between model and expert performance offers an important stepping stone between traditional benchmarks and clinical deployment, helping identify where models need improvement before being tested in real clinical settings.

Methods

BENCHMARK DEVELOPMENT

We developed a publicly available benchmark comprising 750 SCT questions from 10 international medical datasets spanning diverse specialties and clinical domains (Table 1).¹⁷⁻²³ Each SCT test was validated by expert reference panels. Detailed human performance data were available for several of these tests, allowing direct comparisons. One dataset (72 questions) was previously published. The included physiotherapy SCT broadens clinical reasoning evaluation to nonphysician practitioners.

All tests were standardized (Fig. 1B), and performance data were sourced from test contributors (Tests 4-7, Table 1) or relevant publications (Test 3 and Tests 8-10). Inter-rater agreement metrics, where available, are reported in Table S1 in the Supplementary Appendix.

MODEL EVALUATION

We evaluated 10 leading LLMs, selected due to strong performance on other benchmarks: OpenAI’s GPT-4o

A

Diagnosis SCT						
A 27-year-old male presents to the doctor with weakness affecting his right arm. He has a manually repetitive job and also suffered a shoulder dislocation while playing sport 1 week ago.						
If you were thinking of:	And then you find:	This diagnosis becomes:				
Carpal tunnel syndrome	He also complains of “shooting” pain in his neck	Much less likely	Slightly less likely	Neither more nor less likely	Slightly more likely	Much more likely
Investigation SCT						
A 59-year-old female presents to the emergency department with fatigue. She is found to have an elevated bilirubin and elevated ALT/AST. Her liver edge is palpable and on percussion the liver is enlarged.						
If you were thinking of:	And then you find:	This management becomes:				
Ordering an abdominal CT	She has elevated levels of AFP	Much less appropriate	Slightly less appropriate	Neither more nor less appropriate	Slightly more appropriate	Much more appropriate
Treatment SCT						
A 66-year-old male presents to the GP with fatigue for the past 2–4 weeks. Physical examination reveals pallor of his conjunctivae and palmar creases. He has been a vegetarian for many years.						
If you were thinking of:	And then you find:	This management becomes:				
Treating with B ₁₂ supplementation	His MCV is 75 (80–100)	Much less appropriate	Slightly less appropriate	Neither more nor less appropriate	Slightly more appropriate	Much more appropriate

B

Clinical Stem	If you were thinking of:	And then you find:	Category	–2 Score	–1 Score	0 Score	1 Score	2 Score
A 27-year-old male presents to the doctor with weakness affecting his right arm. He has a manually repetitive job and also suffered a shoulder dislocation while playing sport 1 week ago.	Carpal tunnel syndrome	He also complains of “shooting” pain in his neck	Expert opinions	10	7	0	0	0
			Score values	1.0	0.7	0	0	0
A 59-year-old female presents to the emergency department with fatigue. She is found to have an elevated level of bilirubin and elevated levels of ALT/AST. Her liver edge is palpable and on percussion the liver is enlarged.	Ordering an abdominal CT	She has elevated levels of AFP	Expert opinions	0	0	0	1	16
			Score values	0	0	0	0.06	1.0
A 66-year-old male presents to the GP with fatigue for the past 2–4 weeks. Physical examination reveals pallor of his conjunctivae and palmar creases. He has been a vegetarian for many years.	Treating with B ₁₂ supplementation	His MCV is 75 (80–100)	Expert opinions	10	7	0	0	0
			Score values	1.0	0.7	0	0	0

Figure 1. Example of a Script Concordance Test with Scoring Schema.

Panel A demonstrates formatting for students, and Panel B demonstrates formatting for large language model use, as well as an example of the scoring structure. AFP denotes alpha-fetoprotein; ALT, alanine aminotransferase; AST, aspartate transaminase; CT, computed tomography; GP, general practitioner; MCV, mean corpuscular volume; and SCT, script concordance testing.

(gpt-4o-2024-08-06),²⁴ o1-preview (o1-2024-12-17),²⁵ o4-mini (o4-mini-2025-04-16)²⁶, and o3 (o3-2025-04-16);²⁶ Anthropic’s Claude 3.5 Sonnet (claude-3-5-sonnet-latest);²⁷ Google’s Gemini 1.5 Pro (gemini-1.5-pro-002)²⁸

and Gemini 2.5 (gemini-2.5-pro-preview-03-25);²⁹ DeepSeek’s DeepSeek R1;³⁰ Meta’s Llama 3.3 70B (llama-3.3-70b-instruct);³¹ and Mistral AI’s Mistral Small 24B (mistral-small-3.1-24b-instruct) (Mistral Small).³²

Test	Name	Source Institution	Profession	Specialty	Benchmark Comparison	Detailed Response Data?	Publicly Available?	Questions (Total: 750)
1	Open Medical SCT ¹⁷	University of Florida	Physician	Pediatrics	No	No	Yes	72
2	Adelaide SCT*	University of Adelaide	Physician	Medicine, surgery, psychiatry, pediatrics, obstetrics	No	No	No	102
3	Infant LP SCT ¹⁸	Children's Hospital at Montefiore	Physician	Pediatrics — infant LP	226 Attending physicians	No	No	32
4	IU SCT-EM ¹⁹	Indiana University	Physician	Emergency medicine	314 Fourth-year medical students, 40 residents, and 13 attending physicians	Yes	No	59
5	IU Student SCT ²⁰	Indiana University	Physician	Emergency medicine	411 Second-year medical students, 70 fourth-year medical students, and 30 attending physicians	Yes	No	75
6	IU National SCT*	Indiana University	Physician	Emergency medicine	171 Fourth-year medical students	Yes	No	100
7	Neurology SCT ¹⁵	McGill University	Physician	Neurology	8 Medical students, 41 resident physicians, and 16 attending physicians	Yes	No	94
8	Physiotherapy SCT ²¹	University of Indianapolis	Physiotherapist	General physiotherapy	44 First-year DPT students and 15 experienced physiotherapists	No	No	90
9	Singapore IM SCT ²²	Singapore Health Services	Physician	General internal medicine	33 Interns, 22 medical officers, and 20 senior residents	No	No	73
10	Singapore Neurology SCT ²³	National Neuroscience Institute	Physician	Neurology	52 Medical students, 37 resident physicians, and 15 attending physicians	No	No	53

*Not previously published. DPT denotes Doctor of Physical Therapy; EM, emergency medicine; IM, internal medicine; IU, Indiana University; LP, lumbar puncture; and SCT, script concordance testing.

The evaluation pipeline tested each model under multiple conditions: zero-shot inference, in which models were given only basic instructions; few-shot inference, where models were shown five example questions with answers (from the University of Florida's public dataset); and both conditions with and without prompting for explanatory

reasoning.¹⁷ All models were used at default settings, including temperature. Although more sophisticated strategies, such as retrieval-augmented generation,³³ dynamic chain-of-thought³⁴ scaffolding, or Bayesian reasoning prompts,³⁵ could potentially enhance performance, these strategies were beyond the scope of this initial benchmark.

The system prompt remained at default settings across all conditions, while the user prompt contained the guidelines outlining the SCT format and scoring criteria, example questions, and answers in few-shot conditions, and the test question itself. Complete prompting details are provided in Figure S1. All results are reported in line with the TRIPOD-LLM³⁶ guidelines (see the Supplementary Appendix).

STATISTICAL ANALYSES

Aggregate model performance across prompting conditions was compared using repeated-measures Friedman tests with Conover post hoc tests (Bonferroni corrected). Response distributions were compared with Kruskal–Wallis tests and Dunn’s tests using SciPy³⁷ and scikit-learn³⁸ packages.

BENCHMARK AVAILABILITY AND USAGE

The benchmark and leaderboard are publicly accessible via a secure platform (<https://www.concor.dance/>). Two SCT datasets (Open Medical and Adelaide) are publicly available for model fine-tuning. Researchers may submit models for evaluation on the remaining tests upon completing a data use agreement. Results are posted to an online leaderboard comparing models with human clinicians.

Results

OVERVIEW OF TEST IMPLEMENTATION

All included models successfully completed the benchmark, with parseable response rates varying between 96.2 and 99.7%, except for DeepSeek R1, which had a rate as low as 87% in one trial. The overall response rate was 94.7% (Table S2). Performance varied slightly by prompting strategy (Fig. S2).

MODEL PERFORMANCE COMPARISON

Averaged across prompting conditions, OpenAI’s o3 achieved the highest performance ($67.8\% \pm 1.2\%$ SE), followed by GPT-4o ($63.9\% \pm 1.3\%$) (Fig. 2A). Other reasoning-optimized models performed lower, with o1-preview at $58.2\% \pm 1.3\%$ and DeepSeek R1 at $55.5\% \pm 1.4\%$. Among the general-purpose models, Claude 3.5 Sonnet ($59.7\% \pm 1.2\%$) performed comparably to o1-preview, while Gemini 1.5 Pro ($55.4\% \pm 1.4\%$), Mistral Small ($56.1\% \pm 1.3\%$), and Llama 3.3 70B ($57.2\% \pm 1.4\%$) clustered at a poorer level. Gemini 2.5 exhibited the lowest overall mean score ($52.1\% \pm 1.4\%$). Friedman tests

confirmed significant differences ($P < 0.001$). Post hoc Conover tests identified 22 significant pairwise differences (Fig. 2B). OpenAI’s o3 significantly outperformed all other models ($P < 0.05$, Bonferroni corrected), and GPT-4o significantly outperformed all except o3, o4-mini, and Claude 3.5 Sonnet.

PER-TEST PERFORMANCE

Model performance varied substantially across tests (Fig. 3). Between all tests, comparison data were available for 300 attending physicians, 193 resident physicians, 15 staff physiotherapists, 1026 medical students, and 44 physiotherapy students. Although models matched or exceeded student performance on several examinations, they consistently performed below the level of senior residents and attending physicians (aside from o3 on the Singapore Neurology SCT test). All tests, including multiple levels of human clinician skill, show a clear positive performance gradient with expertise, underscoring the face validity of SCT as an assessment of clinical skill.

MODEL RESPONSE DISTRIBUTION ANALYSIS

Models showed distinct deviations from the optimal distribution (Fig. 4). All models underused neutral (0) ratings, indicating no probability adjustment from the given information, with Mistral Small closest to optimal. The reasoning models, in particular Gemini 2.5 and DeepSeek R1, and to a lesser extent o4-mini and o1-preview, demonstrate a substantial bias toward -2 and $+2$ responses. Model o3 maintained a more balanced but still suboptimal distribution.

DETAILED COMPARISON WITH STUDENT RESPONSES

Direct comparisons (Fig. S4) revealed weak correlations between question difficulty for models versus students, particularly for reasoning-focused models other than o3. Models often found questions easy that students found challenging, and vice versa. Student scores typically clustered near perfect scores, while model scores were more evenly distributed. All model response distributions significantly differed from student response distributions.

Discussion

Our study demonstrates that SCT represents a distinctive and challenging benchmark for evaluating clinical reasoning in LLMs. Unlike many existing medical benchmarks where LLMs have achieved superhuman performance,⁹

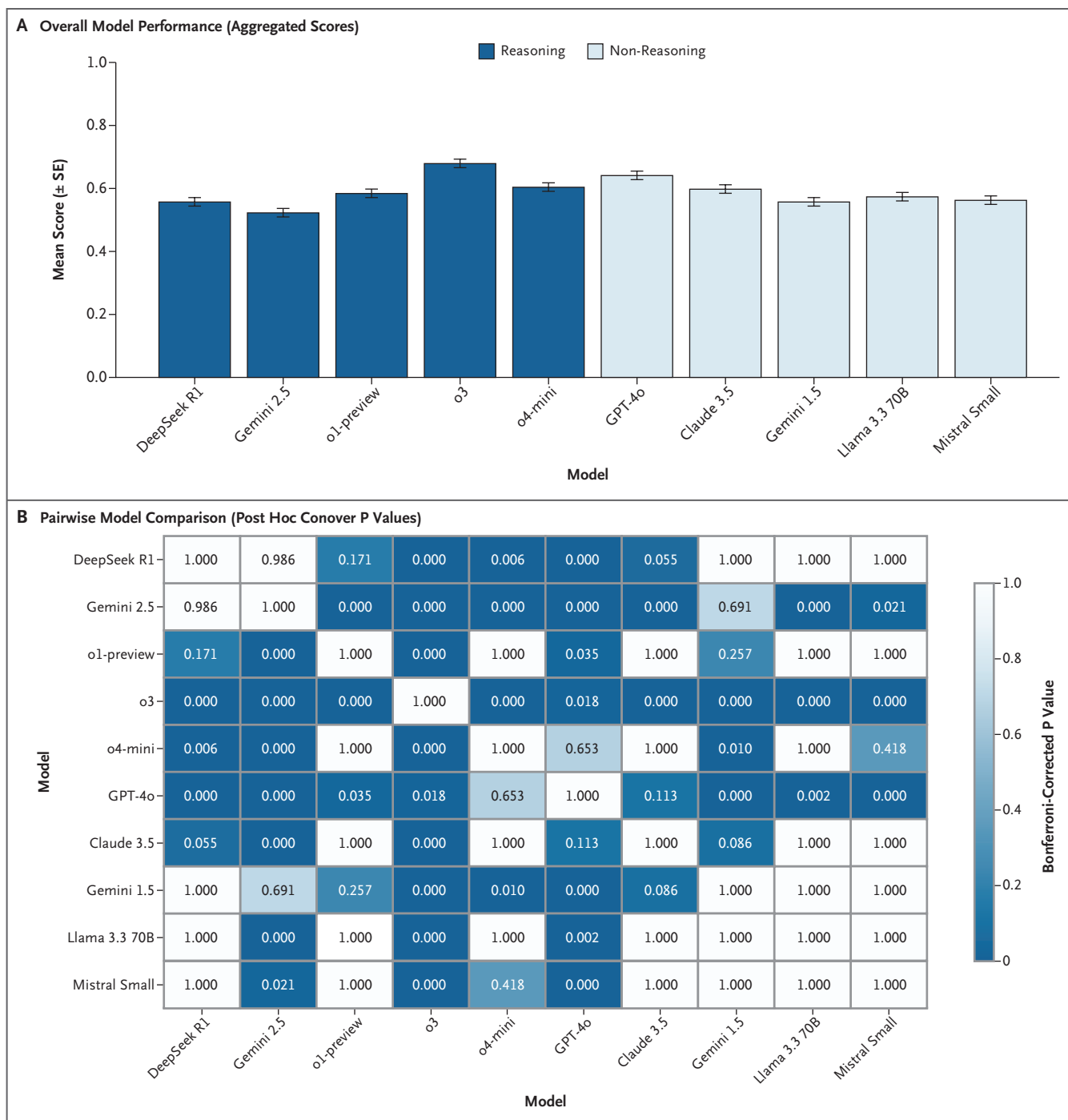


Figure 2. Overall Performance of Models on Full Script Concordance Testing Benchmark.

Scores reported as mean percentage of maximum possible score, $\pm$ SE. Statistical analysis performed with repeated measures Friedman test with post hoc Conover test with Bonferroni correction applied. SE denotes standard error.

SCT reveals significant gaps between model and expert clinician performance, even for the highest performing models. This finding is particularly notable given recent

concerns about benchmark saturation³⁹ in medical artificial intelligence (AI) evaluation. In making this benchmark publicly available, we hope to enable novel work for the

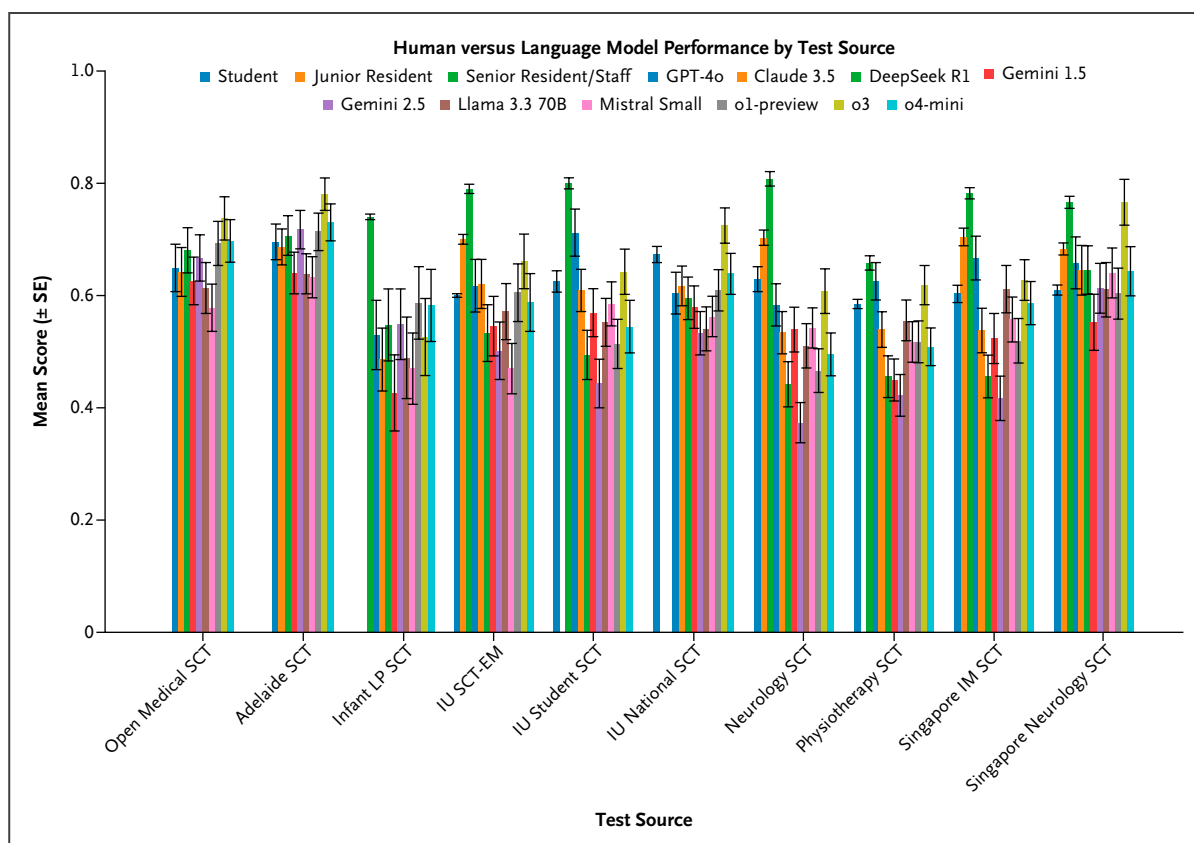


Figure 3. Per-Test Performance of Models and Humans of Varying Training Levels.

Results shown as the mean ratio relative to the maximum possible score, $\pm$ SE. IM denotes internal medicine; IU, Indiana University; LP, lumbar puncture; SCT, script concordance testing; and SE, standard error.

improvement of LLM reasoning both clinically and in more general domains.

Model performance on SCT offers key insights into LLM clinical reasoning. Although models often match or exceed medical students on these tasks, they consistently underperform relative to expert clinicians — a sharp contrast to their success on multiple choice questions (MCQs). This pattern mirrors findings in human learners: in one study, emergency medicine residents demonstrated no significant correlation between SCT and MCQ scores,⁴⁰ suggesting that these formats assess distinct competencies. Another study found that while an MCQ-SCT correlation exists for medical students, this disappears in later years of training,⁴¹ implying that SCT performance increasingly reflects reasoning rather than factual recall. As clinical expertise deepens, success in SCT appears to depend less on encoded knowledge and more on flexible judgment under uncertainty.

Given robust evidence of knowledge encoding in models, increased gains in performance on SCT may suggest improved reasoning abilities. However, it is important to acknowledge that, while valuable, SCT offers a constrained and abstracted representation of clinical reasoning by removing context specificity — the individual, interpersonal, systems, and societal factors that affect clinical reasoning⁴² — a limitation that also applies to their use in the evaluation of humans.

Beyond the overall performance gap, model rankings on SCT diverged from expectations based on standard benchmarks. OpenAI's o3 model, which incorporates explicit reasoning optimization and tool use,²⁶ achieved the highest overall performance on the SCT benchmark; however, its margin was modest compared with its dominance on standard medical benchmarks.⁴³ Moreover, other reasoning-specialized models, including o1-preview, DeepSeek R1, and Gemini 2.5, failed to demonstrate expected

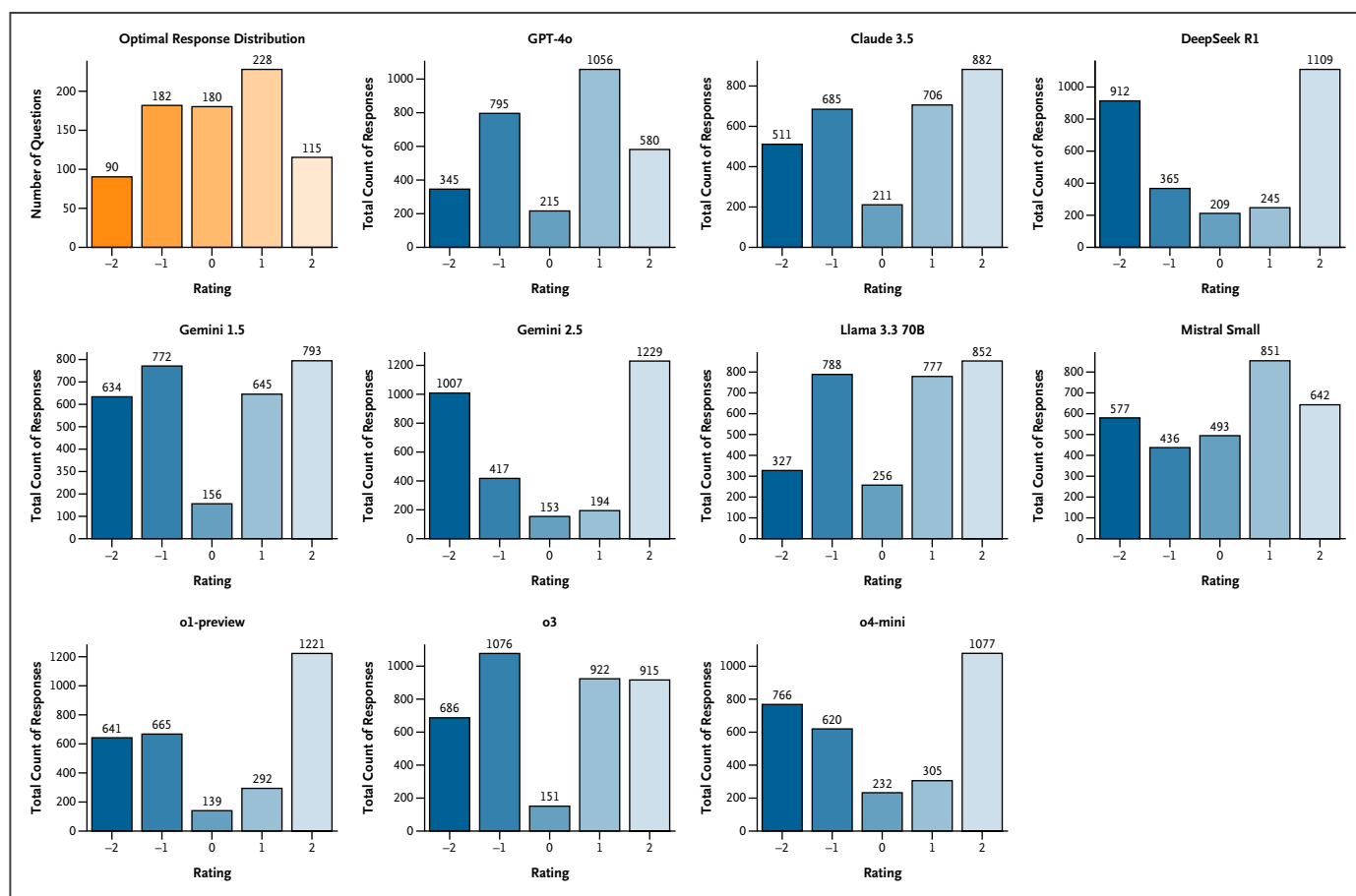


Figure 4. Comparison of Optimal Response Distribution with Overall Model Response Distribution.

Plots denote total sums of model responses, as compared with the ideal response distribution with perfect concordance to the human experts (top-left plot).

performance gains (with the last notably underperforming compared with its own predecessor).

Model response patterns further clarified these shortcomings. Compared with human experts, all models underused the 0 response, failing to recognize when new information should not alter a hypothesis. Reasoning-tuned models also showed a marked bias toward extreme responses (+2/-2), suggesting overconfidence. This tendency toward exaggerated shifts may explain why reasoning-focused architectures struggle with SCT, where success often requires subtle, probabilistic updating rather than assertive recalibration. However, this underperformance may be prompt sensitive, and our findings should be seen as a demonstration of current limitations under default prompting, not as definitive ceilings on model capability.

These results parallel findings from Liu et al. in 2024,¹⁰ who demonstrated that chain-of-thought prompting

systematically reduces model performance on tasks where human verbal deliberation also harms outcomes. In their experiments, tasks requiring implicit statistical learning, exception handling, or subtle pattern recognition showed significant performance drops under chain-of-thought prompting. These findings suggest that structured, step-by-step reasoning may promote overconfidence or premature commitment to incorrect hypotheses in contexts of uncertainty — precisely the kind of environment modeled by SCT. Thus, while reasoning optimization remains beneficial, SCT reveals that its advantages are fragile and context sensitive. The benchmark offers a rare lens on flexible, probabilistic reasoning — complementing traditional, fact-based evaluations. These findings also reflect a broader challenge in evaluating clinical reasoning: the most robust assessments are those validated against expert performance.⁴⁴ Although standardized test scores correlate with patient outcomes,⁴⁵ they are insufficient proxies for clinical

competence. Modern medical education has shifted toward competency-based frameworks⁴⁶ that emphasize workplace-based assessments — stressing the importance of broader, often subjective, assessments, which are difficult to replicate algorithmically.

Recent proposals to simulate entire encounters using LLM-based role-play^{47,48} offer technological sophistication but risk further abstraction from real clinical reasoning. Our results suggest a more grounded alternative: human-validated benchmarks targeting specific, high-value reasoning tasks.⁴⁴ These allow for focused evaluation without presuming to capture the totality of clinical judgment. Ultimately, just as medical education has evolved toward direct observation in practice, LLM evaluation must also progress toward richer, trial-based assessments. Our benchmark contributes to this path by identifying specific reasoning limitations that can inform both model development and future trial design.

Our findings must be interpreted in light of both the benchmark's strengths and its inherent constraints. A major strength lies in its diversity: our SCT benchmark spans North America, Asia, and Australia, covering a range of generalist and specialist contexts. This geographic and clinical breadth enhances the benchmark's ecological validity, particularly as reasoning patterns and clinical heuristics often vary by setting. That LLM performance remained consistently below expert levels across these diverse contexts highlights both the robustness of the SCT format and the limitations of current models. Moreover, several SCT tests include responses across multiple stages of medical training, enabling comparisons that align closely with real-world learning trajectories. Importantly, with only one exception, all SCT questions were previously unpublished, eliminating the risk of data leakage.

Still, key limitations remain. As with all LLM assessments, SCT performance is prompt and context sensitive. Our evaluation used standard prompting approaches; future work employing optimized prompts,⁴⁹ Bayesian frameworks,³⁵ or tool-augmented reasoning may yield different outcomes. We welcome further engagement with the public benchmark to better explore these approaches and determine the degree to which these results may be sensitive to prompt engineering. This caveat extends to both the aggregate findings and the interpretation of model-level differences, including claims regarding the performance of reasoning-specialized models.

In addition, the SCT format itself introduces potential scoring artifacts: expert panels tend to avoid extreme responses,⁵⁰ potentially inflating scores for models that mirror this conservatism. The assumption that all expert

opinions have equal validity also risks partial credit for objectively incorrect answers.⁵¹ Moreover, inconsistent interpretation of the Likert scale — where some interpret extremes as markers of confidence, others as probability shifts — may cloud both human and model scoring.⁵²

Furthermore, while geographic and institutional diversity strengthens generalizability, it also complicates interpretation. Clinical reasoning is highly context specific,⁵³ and expert consensus in one setting may not hold in another. Similarly, the inclusion of physiotherapy SCT introduces reasoning patterns not directly comparable with physician practice. Our use of aggregate performance also masks variations across prompting strategies. Future work should investigate these variations more systematically and explore how different prompting strategies might better align with the cognitive processes SCT aims to evaluate.

Finally, our results spotlight a foundational challenge in AI evaluation: how should models be judged as they approach — or surpass — human performance? In real clinical practice, expert disagreement is common and often appropriate. The SCT scoring method, which rewards concordance with expert distributions, may inadvertently penalize genuinely superior reasoning that departs from consensus. A superhuman model might justifiably diverge from human patterns yet be scored lower for doing so. This paradox underscores the epistemic limits of concordance-based metrics.

Despite these limitations, SCT remains a uniquely valuable tool. It stresses probabilistic, context-sensitive reasoning under uncertainty — capacities essential for clinical decision-making and inadequately captured by existing benchmarks.⁵⁴ As such, it offers a crucial test bed for model development, especially in domains where subtle judgment matters more than factual recall.

Future work should extend SCT-based evaluation across health professions (e.g., pharmacy, nursing, and additional physiotherapy domains) and collect item-level rater data to enable inter-rater agreement analyses. Ultimately, just as clinical education evolved toward direct observation of real-world performance, so too must LLM evaluation move toward richer, trial-based testing. Our benchmark lays important groundwork for this progression by surfacing where today's models still fall short.

Conclusion

Our SCT benchmark reveals important limitations in LLM clinical reasoning capabilities that are not apparent

in traditional benchmarks. Although current models can match or exceed medical student performance on many SCT tests, they consistently fall short of senior resident and expert clinician performance. The unexpected performance patterns observed among models — particularly the underperformance of reasoning-optimized models relative to expectations from prior benchmarks — underscore that explicit reasoning enhancements do not uniformly improve clinical reasoning performance.

Our analysis of model response patterns reveals that even state-of-the-art LLMs exhibit striking overconfidence, disproportionately favoring extreme belief shifts and failing to recognize when new information should not alter clinical hypotheses. By challenging models to make flexible, uncertainty-sensitive adjustments, SCT provides a valuable complement to conventional evaluations and offers an important benchmark for next-generation clinical AI.

These results highlight broader challenges in evaluating AI systems as they approach human-level performance in medicine, particularly in domains where expert disagreement is both common and clinically appropriate. SCT's concordance-based scoring system, which acknowledges legitimate disagreement among experts, offers a more sophisticated framework for assessing clinical reasoning than traditional right/wrong evaluations. However, this approach also raises important questions about how to evaluate AI systems that may potentially exceed human consensus in certain areas, particularly as these systems move closer to clinical deployment.

To facilitate ongoing development, we are making portions of our benchmark publicly available for model development, with the remainder accessible through a secure evaluation framework. This tiered access approach balances the need for open science and reproducibility with test integrity. By providing both training data and held-out test sets, we enable researchers to develop and validate new approaches while ensuring meaningful comparisons across different models and methodologies. As the field of medical AI continues to evolve rapidly, maintaining such standardized evaluation frameworks — particularly those validated against human expert performance — will be crucial not only for tracking technical progress, but ultimately for ensuring that AI systems can safely and effectively support clinical decision-making in real-world health care settings.

Disclosures

Author disclosures and other supplementary materials are available at ai.nejm.org.

In all cases where detailed student data were used, this is covered by the original institutional review boards for the component datasets. For Indiana University script concordance tests 4–6, the institutional review board of the Indiana University School of Medicine gave ethics approval. For neurology script concordance test 7, the McGill University Institutional Ethics Board gave ethics approval. For all other script concordance tests, any human performance data used in this article were extracted in summary form from the openly published final articles as cited in [Table 1](#), which were open to the public prior to the initiation of the study.

Given the privacy agreements with the various research groups that have agreed to provide these tests, data access will be available in the form of a hosted benchmark available for model testing upon signature of a data use agreement (<https://concor.dance>).

Supported by the Fulbright Program, the Gordon and Betty Moore Foundation, and the Josiah Macy Jr. Foundation.

We thank the authors of the script concordance tests who graciously agreed to provide us with access to their prior work. In addition, we wish to acknowledge Stuart Lubarsky, Lindsey Kojich, Stephanie A. Miller, Todd P. Chang, Martin V. Pusic, Nnenna O. Chime, Peggy Seidman, Mary Johnson, Bart Besinger, and Pranshu Bhardwaj.

Author Affiliations

- ¹ University of Alberta, Edmonton
- ² Beth Israel Deaconess Medical Center, Boston
- ³ Massachusetts Institute of Technology, Cambridge
- ⁴ Harvard University, Cambridge, MA
- ⁵ University of Adelaide, Adelaide, SA, Australia
- ⁶ Sengkang General Hospital, Singapore
- ⁷ National Neuroscience Institute, Singapore
- ⁸ Duke-NUS Medical School, Singapore
- ⁹ Department of Biomedical Informatics, Harvard Medical School, Boston
- ¹⁰ Harvard T.H. Chan School of Public Health, Harvard University, Boston
- ¹¹ NEJM AI, Boston
- ¹² Indiana University School of Medicine, Indianapolis

References

1. Albrecht M, Shanks D, Shah T, et al. Enhancing clinical documentation with ambient artificial intelligence: a quality improvement survey assessing clinician perspectives on work burden, burnout, and job satisfaction. *JAMIA Open* 2025;8:ooaf013. DOI: [10.1093/jamiaopen/ooaf013](https://doi.org/10.1093/jamiaopen/ooaf013).
2. McCoy LG, Manrai AK, Rodman A. Large language models and the degradation of the medical record. *N Engl J Med* 2024;391:1561–1564. DOI: [10.1056/NEJMp2405999](https://doi.org/10.1056/NEJMp2405999).
3. Duggan MJ, Gervase J, Schoenbaum A, et al. Clinician experiences with ambient scribe technology to assist with documentation burden and efficiency. *JAMA Netw Open* 2025;8:e2460637. DOI: [10.1001/jamanetworkopen.2024.60637](https://doi.org/10.1001/jamanetworkopen.2024.60637).

4. Liu S, Wright AP, McCoy AB, et al. Using large language model to guide patients to create efficient and comprehensive clinical care message. *J Am Med Inform Assoc* 2024;31:1665-1670. DOI: [10.1093/jamia/ocae142](https://doi.org/10.1093/jamia/ocae142).
5. Van Veen D, Van Uden C, Blankemeier L, et al. Adapted large language models can outperform medical experts in clinical text summarization. *Nat Med* 2024;30:1134-1142. DOI: [10.1038/s41591-024-02855-5](https://doi.org/10.1038/s41591-024-02855-5).
6. Ke YH, Jin L, Elangovan K, et al. Real-world deployment and evaluation of PEriooperative AI CHatbot (PEACH) — a large language model chatbot for perioperative medicine. December 24, 2024 (<https://doi.org/10.48550/arXiv.2412.18096>). Preprint.
7. Han S, Choi W. Development of a large language model-based multi-agent clinical decision support system for Korean triage and acuity scale (KTAS)-based triage and treatment planning in emergency departments. August 27, 2024 (<https://doi.org/10.48550/arXiv.2408.07531>). Preprint.
8. Griot M, Vanderdonckt J, Yuxsel D, Hemptinne C. Pattern recognition or medical knowledge? The problem with multiple-choice questions in medicine. June 4, 2024 (<https://doi.org/10.48550/arXiv.2406.02394>). Preprint.
9. Brodeur PG, Buckley TA, Kanjee Z, et al. Superhuman performance of a large language model on the reasoning tasks of a physician. June 2, 2025 (<https://doi.org/10.48550/arXiv.2412.10849>). Preprint.
10. Liu R, Geng J, Wu AJ, Sucholutsky I, Lombrozo T, Griffiths TL. Mind your step (by step): chain-of-thought can reduce performance on tasks where thinking makes humans worse. November 8, 2024 (<https://doi.org/10.48550/arXiv.2410.21333>). Preprint.
11. Tu T, Palepu A, Schaekermann M, et al. Towards conversational diagnostic AI. January 11, 2024 (<https://doi.org/10.48550/arXiv.2401.05654>). Preprint.
12. Goh E, Gallo R, Hom J, et al. Large language model influence on diagnostic reasoning: a randomized clinical trial. *JAMA Netw Open* 2024;7:e2440969. DOI: [10.1001/jamanetworkopen.2024.40969](https://doi.org/10.1001/jamanetworkopen.2024.40969).
13. Cabral S, Restrepo D, Kanjee Z, et al. Clinical reasoning of a generative artificial intelligence model compared with physicians. *JAMA Intern Med* 2024;184:581-583. DOI: [10.1001/jamainternmed.2024.0295](https://doi.org/10.1001/jamainternmed.2024.0295).
14. Strong E, DiGiammarino A, Weng Y, et al. Chatbot versus medical student performance on free-response clinical reasoning examinations. *JAMA Intern Med* 2023;183:1028-1030. DOI: [10.1001/jamainternmed.2023.2909](https://doi.org/10.1001/jamainternmed.2023.2909).
15. Lubarsky S, Charlin B, Cook DA, Chalk C, van der Vleuten CPM. Script concordance testing: a review of published validity evidence. *Med Educ* 2011;45:329-338. DOI: [10.1111/j.1365-2923.2010.03863.x](https://doi.org/10.1111/j.1365-2923.2010.03863.x).
16. Lubarsky S, Dory V, Duggan P, Gagnon R, Charlin B. Script concordance testing: from theory to practice: AMEE guide no. 75. *Med Teach* 2013;35:184-193. DOI: [10.3109/0142159X.2013.760036](https://doi.org/10.3109/0142159X.2013.760036).
17. Bhardwaj P, Black EW, Fantone JC, Lopez M, Kelly M. Script concordance tests for formative clinical reasoning and problem-solving assessment in general pediatrics. *MedEdPORTAL* 2022;18:11274. DOI: [10.15766/mep_2374-8265.11274](https://doi.org/10.15766/mep_2374-8265.11274).
18. Chime NO, Pusic MV, Auerbach M, et al. Script concordance testing to determine infant lumbar puncture practice variation. *Pediatr Emerg Care* 2018;34:84-92. DOI: [10.1097/PEC.0000000000000851](https://doi.org/10.1097/PEC.0000000000000851).
19. Humbert AJ, Besinger B, Miech EJ. Assessing clinical reasoning skills in scenarios of uncertainty: convergent validity for a Script Concordance Test in an emergency medicine clerkship and residency. *Acad Emerg Med* 2011;18:627-634. DOI: [10.1111/j.1553-2712.2011.01084.x](https://doi.org/10.1111/j.1553-2712.2011.01084.x).
20. Humbert AJ, Johnson MT, Miech E, Friedberg F, Grackin JA, Seidman PA. Assessment of clinical reasoning: a script concordance test designed for pre-clinical medical students. *Med Teach* 2011;33:472-477. DOI: [10.3109/0142159X.2010.531157](https://doi.org/10.3109/0142159X.2010.531157).
21. Kojich L, Miller SA, Axman K, Eacret T, Koontz JA, Smith C. Evaluating clinical reasoning in first year DPT students using a script concordance test. *BMC Med Educ* 2024;24:329. DOI: [10.1186/s12909-024-05281-w](https://doi.org/10.1186/s12909-024-05281-w).
22. Fong JMN, Hoe RHM, Huang DH, et al. Script concordance test to assess diagnostic and management reasoning in acute medicine. *Ann Acad Med Singap* 2023;52:383-385. DOI: [10.47102/annals-acadmedsg.202327](https://doi.org/10.47102/annals-acadmedsg.202327).
23. Tan K, Tan NCK, Kandiah N, Samarasekera D, Ponnampereuma G. Validating a script concordance test for assessing neurological localization and emergencies. *Eur J Neurol* 2014;21:1419-1422. DOI: [10.1111/ene.12373](https://doi.org/10.1111/ene.12373).
24. OpenAI. Hello GPT-4o. May 13, 2024. <https://openai.com/index/hello-gpt-4o/>.
25. OpenAI. Introducing OpenAI o1-preview. September 12, 2024. <https://openai.com/index/introducing-openai-o1-preview/>.
26. OpenAI. Introducing OpenAI o3 and o4-mini. September 17, 2024. <https://openai.com/index/introducing-o3-and-o4-mini/>.
27. Anthropic. Introducing computer use, a new Claude 3.5 Sonnet, and Claude 3.5 Haiku. October 22, 2024. <https://www.anthropic.com/news/3-5-models-and-computer-use>.
28. Google. Our next-generation model: Gemini 1.5. February 15, 2024. <https://blog.google/technology/ai/google-gemini-next-generation-model-february-2024/>.
29. Google. Gemini 2.5: our most intelligent AI model. March 25, 2025. <https://blog.google/technology/google-deepmind/gemini-model-thinking-updates-march-2025/>.
30. DeepSeek-AI, Guo D, Yang D, et al. DeepSeek-R1: incentivizing reasoning capability in LLMs via reinforcement learning. January 22, 2025 (<https://doi.org/10.48550/arXiv.2501.12948>). Preprint.

31. Grattafiori A, Dubey A, Jauhri A, et al. The Llama 3 herd of models. November 23, 2024 (<https://doi.org/10.48550/arXiv.2407.21783>). Preprint.
32. Mistral AI. Mistral Small 3.1. March 17, 2025. <https://mistral.ai/news/mistral-small-3-1>.
33. Xiong G, Jin Q, Wang X, Zhang M, Lu Z, Zhang A. Improving retrieval-augmented generation in medicine with iterative follow-up questions. October 11, 2024 (<https://doi.org/10.48550/arXiv.2408.00727>). Preprint.
34. Wei J, Wang X, Schuurmans D, et al. Chain-of-thought prompting elicits reasoning in large language models. January 10, 2023 (<https://doi.org/10.48550/arXiv.2201.11903>). Preprint.
35. Ellis K. Human-like few-shot learning via Bayesian reasoning over natural language. September 29, 2023 (<https://doi.org/10.48550/arXiv.2306.02797>). Preprint.
36. Gallifant J, Afshar M, Ameen S, et al. The TRIPOD-LLM reporting guideline for studies using large language models. *Nat Med* 2025;31:60-69. DOI: [10.1038/s41591-024-03425-5](https://doi.org/10.1038/s41591-024-03425-5).
37. Virtanen P, Gommers R, Oliphant TE, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat Methods* 2020;17:261-272. DOI: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2).
38. Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: machine learning in Python. *J Mach Learn Res* 2011;12:2825-2830.
39. Ott S, Barbosa-Silva A, Blagec K, Brauner J, Samwald M. Mapping global dynamics of benchmark creation and saturation in artificial intelligence. *Nat Commun* 2022;13:6793. DOI: [10.1038/s41467-022-34591-0](https://doi.org/10.1038/s41467-022-34591-0).
40. Fournier J-P, Thiercelin D, Pulcini C, et al. Clinical reasoning assessment in emergency medicine: script concordance tests are more efficient to detect clinical experience than rich-context multiple-choice questions. *Pédagogie Médicale* 2006;7:20-30. DOI: [10.1051/pmed:2006020](https://doi.org/10.1051/pmed:2006020).
41. Collard A, Gelaes S, Vanbelle S, et al. Reasoning versus knowledge retention and ascertainment throughout a problem-based learning curriculum. *Med Educ* 2009;43:854-865. DOI: [10.1111/j.1365-2923.2009.03410.x](https://doi.org/10.1111/j.1365-2923.2009.03410.x).
42. Konopasky A, Artino AR, Battista A, et al. Understanding context specificity: the effect of contextual factors on clinical reasoning. *Diagn Berl Ger* 2020;7:257-264. DOI: [10.1515/dx-2020-0016](https://doi.org/10.1515/dx-2020-0016).
43. OpenAI. Introducing HealthBench. May 12, 2025. <https://openai.com/index/healthbench/>.
44. Rodman A, Zwaan L, Olson A, Manrai AK. When it comes to benchmarks, humans are the only way. *NEJM AI* 2025;2(4). DOI: [10.1056/AIe2500143](https://doi.org/10.1056/AIe2500143).
45. Gray BM, Vandergrift JL, Stevens JP, Lipner RS, McDonald FS, Landon BE. Associations of internal medicine residency milestone ratings and certification examination scores with patient outcomes. *JAMA* 2024;332:300-309. DOI: [10.1001/jama.2024.5268](https://doi.org/10.1001/jama.2024.5268).
46. Frank JR, Snell LS, Cate OT, et al. Competency-based medical education: theory to practice. *Med Teach* 2010;32:638-645. DOI: [10.3109/0142159X.2010.501190](https://doi.org/10.3109/0142159X.2010.501190).
47. Schmidgall S, Ziaei R, Harris C, Reis E, Jopling J, Moor M. AgentClinic: a multimodal agent benchmark to evaluate AI in simulated clinical environments. October 20, 2024 (<https://doi.org/10.48550/arXiv.2405.07960>). Preprint.
48. Johri S, Jeong J, Tran BA, et al. An evaluation framework for clinical use of large language models in patient interaction tasks. *Nat Med* 2025;31:77-86. DOI: [10.1038/s41591-024-03328-5](https://doi.org/10.1038/s41591-024-03328-5).
49. Nori H, Lee YT, Zhang S, et al. Can generalist foundation models outcompete special-purpose tuning? Case study in medicine. November 27, 2023 (<https://doi.org/10.48550/arXiv.2311.16452>). Preprint.
50. Lineberry M, Kreiter CD, Bordage G. Threats to validity in the use and interpretation of script concordance test scores. *Med Educ* 2013;47:1175-1183. DOI: [10.1111/medu.12283](https://doi.org/10.1111/medu.12283).
51. Lineberry M, Kreiter CD, Bordage G. Script concordance tests: strong inferences about examinees require stronger evidence. *Med Educ* 2014;48:452-453. DOI: [10.1111/medu.12417](https://doi.org/10.1111/medu.12417).
52. Lurie S. Towards greater clarity in the role of ambiguity in clinical reasoning. *Med Educ* 2011;45:326-328. DOI: [10.1111/j.1365-2923.2011.03938.x](https://doi.org/10.1111/j.1365-2923.2011.03938.x).
53. Boyle JG, Walters MR, Burton FM, et al. On context specificity and management reasoning: moving beyond diagnosis. *Diagn Berl Ger* 2025;12:217-222. DOI: [10.1515/dx-2024-0122](https://doi.org/10.1515/dx-2024-0122).
54. Raji ID, Daneshjou R, Alsentzer E. It's time to bench the medical exam benchmark. *NEJM AI* 2025;2(2). DOI: [10.1056/AIe2401235](https://doi.org/10.1056/AIe2401235).